# Lakshya JEE 2027

## JEE Replica Sheet [Mains & Advance]

## Physics

## Electrostatics

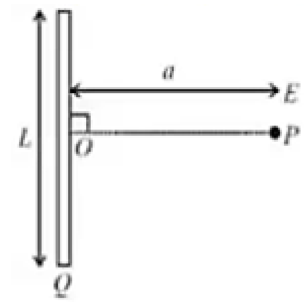
**Q1** Two opposite and equal charges $4 \times 10^{-8}$ coulomb when placed $2 \times 10^{-2}$ cm away, form a dipole. If this dipole is placed parallel to an external electric field of $4 \times 10^{8}$ newton/coulomb, the value of maximum torque and the work done in rotating it through $180^{\circ}$ will be:

(A) $64 \times 10^{-4}$ Nm and $64 \times 10^{-4}$ J
(B) $32 \times 10^{-4}$ Nm and $32 \times 10^{-4}$ J
(C) $64 \times 10^{-4}$ Nm and $34 \times 10^{-4}$ J
(D) $32 \times 10^{-4}$ Nm and $64 \times 10^{-4}$ J

**Q2** An electric dipole is situated in an electric field of uniform intensity $E$ whose dipole moment is $p$ and moment of inertia is $I$. If the dipole is displaced slightly from the equilibrium position, then the angular frequency of its oscillations is

(A) $\left(\frac{pE}{I}\right)^{1/2}$
(B) $\left(\frac{pE}{I}\right)^{3/2}$
(C) $\left(\frac{I}{pE}\right)^{1/2}$
(D) $\left(\frac{p}{IE}\right)^{1/2}$

**Q3** Find the electric field at point $P$ (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length $L$ carrying a charge $Q$. The distance of the point $P$ from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$.




(A) $\frac{Q}{3\pi\epsilon_0 L^2}$
(B) $\frac{Q}{4\pi\epsilon_0 L^2}$
(C) $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$
(D) $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$

**Q4** An oil drop of radius $2$ mm with a density $3$ g cm$^{-3}$ is held stationary under a constant electric field $3.55 \times 10^{5}$ Vm$^{-1}$ in the Millikans oil drop experiment. What is the number of excess electrons that the oil drop will possess? (consider $g = 9.81$ m/s$^2$ )
[JEE Main 2021, 18 Mar (Shift 1)]

(A) $48.8 \times 10^{11}$
(B) $1.73 \times 10^{10}$
(C) $17.3 \times 10^{10}$
(D) $1.73 \times 10^{12}$

**Q5** A certain charge $Q$ is divided into two parts $q$ and $(Q-q)$. How should the charge $Q$ and $q$ be divided so that $q$ and $(Q-q)$ placed at a certain distance apart experience maximum electrostatic repulsion?

(A) $Q = \frac{q}{2}$
(B) $Q = 2q$

(C) $Q=4q$
(D) $Q=3q$

**Q6** Two identical tennis balls each having mass $m$ and charge $q$ are suspended from a fixed point by threads of length $l$ . What is the equilibrium separation when each thread makes a small angle $\theta$ with the vertical?

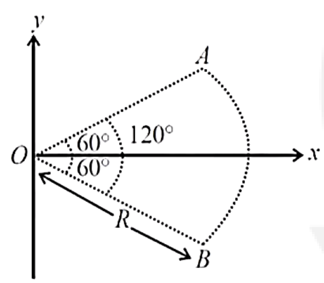
(A) $x=\left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{2}}$

(B) $x=\left(\frac{q^2 l}{2\pi\varepsilon_0 mg}\right)^{\frac{1}{3}}$

(C) $x=\left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g}\right)^{\frac{1}{3}}$

(D) $x=\left(\frac{q^2 l^2}{2\pi\varepsilon_0 m^2 g^2}\right)^{\frac{1}{3}}$

**Q7** Figure shows a rod $AB$, which is bent in a $120^{\circ}$ circular arc of radius $R$. A charge $(-Q)$ is uniformly distributed over rod $AB$. What is the electric field $E$ at the centre of curvature $O$ ?



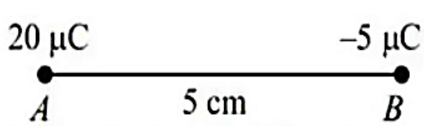

(A) $\frac{3\sqrt{3}Q}{8\pi\varepsilon_0 R^2}(\hat{i})$

(B) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(\hat{i})$

(C) $\frac{3\sqrt{3}Q}{16\pi^2\varepsilon_0 R^2}(\hat{i})$

(D) $\frac{3\sqrt{3}Q}{6\pi^2\varepsilon_0 R^2}(\hat{i})$

**Q8** Two particles A and B having charges $20\mu\mathrm{C}$ and $-5\mu\mathrm{C}$ respectively are held with a separation of $5\ \mathrm{cm}$. At what position a third charged particle should be placed so that it does not experience a net electric force?




(A) At $5\ \mathrm{cm}$ from $20\mu\mathrm{C}$ on the left side of system
(B) At $5\ \mathrm{cm}$ from $-5\mu\mathrm{C}$ on the right side
(C) At $1.25\ \mathrm{cm}$ from $-5\mu\mathrm{C}$ between two charges
(D) At midpoint between two charges

**Q9** Two point charges $+8q$ and $-2q$ are located at $x=0$ and $x=L$ respectively. The location of a point on the $x$ axis at which the net electric field due to these two point charges is zero is

(A) $8L$ (B) $4L$
(C) $2L$ (D) $L/4$

**Q10** Two positive charges of magnitude $q$ are placed at the ends of a side (side 1 ) of a square of side 2a. Two negative charges of the same magnitude are kept at the other corners. Starting from rest, if a charge $Q$ moves from the middle of side 1 to the centre of square, its kinetic energy at the centre of square is:

(A) Zero

(B) $\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{a}\left(1+\frac{1}{\sqrt{5}}\right)$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{a}\left(1-\frac{2}{\sqrt{5}}\right)$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{2qQ}{a}\left(1-\frac{1}{\sqrt{5}}\right)$

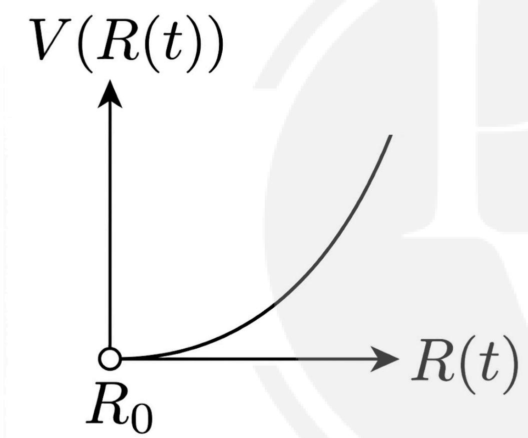
**Q11** Two charges, each equal to $q$, are kept at $x=-a$ and $x=a$ on the $x$-axis. A particle of mass $m$ and charge $q_0=\frac{q}{2}$ is placed at the origin. If charge $q_0$ is given a small displacement $(y<<a)$ along the $y$-axis, the net force acting on the particle is proportional to:

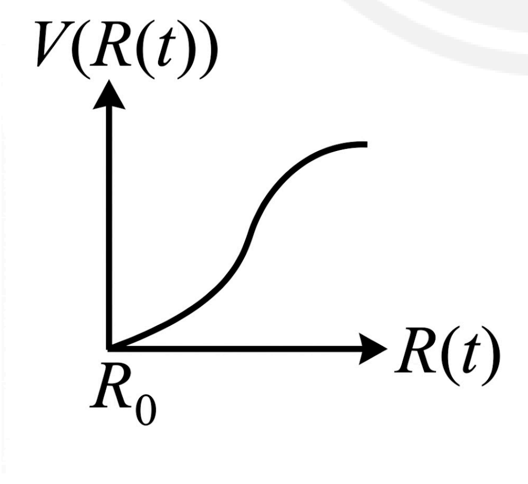
(A) $y$
(B) $-y$
(C) $\frac{1}{y}$
(D) $-\frac{1}{y}$

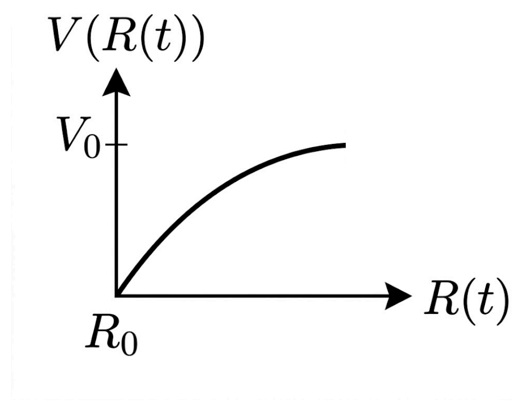
**Q12** An electric field of $1000\ \mathrm{V/m}$ is applied to an electric dipole at angle of $45^{\circ}$. The value of

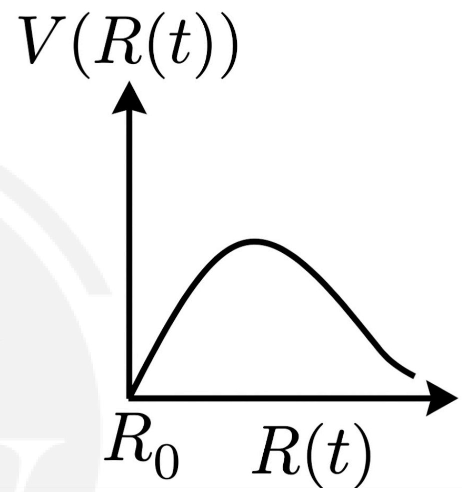
electric dipole moment is $10^{-29}$ C.m. What is the potential energy of the electric dipole?
(A) $-7 \times 10^{-27}$ J
(B) $-20 \times 10^{-18}$ J
(C) $-9 \times 10^{-20}$ J
(D) $-10 \times 10^{-29}$ J

**Q13** There is a uniform spherically symmetric surface charge density at a distance $R_0$ from the origin. The charge distribution is initially at rest and starts expanding because of mutual repulsion. The figure that represents best the speed $V(R(t))$ of the distribution as a function of its instantaneous radius $R(t)$ is:

[JEE Main 2019, 12 Jan (Shift 1)]
(A)


(B)


(C)


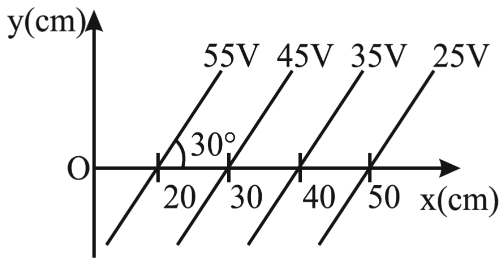
(D)



**Q14** In an arrangement of two concentric conducting shells, with centre at origin and radii $a$ and $b$ $(a < b)$, charges $Q_1$ and $Q_2$ are given to inner and outer shell, then the potential $V$ at a distance $r$ from the origin is $(r < a)$
(A) $V = \frac{kQ_1}{a} + \frac{kQ_2}{b}$
(B) $V = \frac{kQ_1}{r} + \frac{kQ_2}{a}$
(C) $V = \frac{kQ_1}{a} + \frac{kQ_2}{r}$
(D) $V = k\frac{(Q_1+Q_2)}{r}$

**Q15** An infinite number of charges each equal to ' $Q$ ' are placed along the X-axis at $x = 1, x = 2, x = 4, x = 8$ ..... The potential at the point $x = 0$ due to this set of charges is (All quantities are measured in SI units)
(A) $\frac{Q}{4\pi\epsilon_0}$

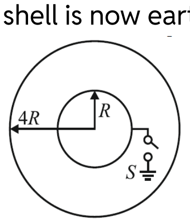
(B) $\frac{2Q}{4\pi\epsilon_0}$
(C) $\frac{3Q}{4\pi\epsilon_0}$
(D) $\frac{Q}{\pi\epsilon_0}$

**Q16** For the given equipotential surfaces, the magnitude and direction of electric field is:




(A) $200 \mathrm{~V/m}, 120^{\circ}$ from positive $x$-axis in anticlockwise direction.
(B) $100 \mathrm{~V/m}, 60^{\circ}$ from positive $x$-axis in anticlockwise direction.
(C) $200 \mathrm{~V/m}, 60^{\circ}$ from positive $x$-axis in clockwise direction.
(D) $100 \mathrm{~V/m}, 120^{\circ}$ from positive $x$-axis in clockwise direction.

**Q17** Two thin conducting shells of radii $R$ and $4R$ are shown in the figure. The outer shell carries a charge
$+2Q$ and the inner shell is initially neutral. The inner
shell is now earthed with the help of switch $S$;



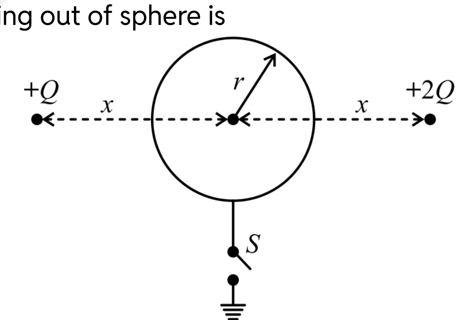

If switch $S$ is closed, the charge attained by the inner sphere is;
(A) $\frac{+Q}{2}$
(B) $\frac{-Q}{2}$
(C) $+Q$
(D) $-Q$

**Q18** A regular hexagon of side 10 cm has a charge $5\ \mu\mathrm{C}$ at each of its vertices. Calculate the potential at the centre of the hexagon.
(A) $3.7 \times 10^6$ V
(B) $2.7 \times 10^6$ V
(C) $2.7 \times 10^4$ V
(D) $3.7 \times 10^4$ V

**Q19** Two unlike charge of magnitude $\mathrm{q}$ are separated by a distance $5\ \mathrm{d}$. The potential at a point midway between them
(A) Zero
(B) $\frac{1}{4\pi\varepsilon_o}\frac{\mathrm{q}}{\mathrm{d}}$
(C) $\frac{3}{2\pi\varepsilon_o}\frac{\mathrm{q}}{\mathrm{d}}$
(D) $\frac{5}{4\pi\varepsilon_o}\frac{\mathrm{q}}{\mathrm{d}}$

**Q20** On moving a charge of $2\mathrm{C}$ from a point $\mathrm{A}$ where potential is $+12\ \mathrm{V}$ to a point $\mathrm{B}$ where potential is
$-12\ \mathrm{V}$, the work done is
(A) $+24\ \mathrm{J}$
(B) $-24\ \mathrm{J}$
(C) $48\ \mathrm{J}$
(D) $-48\ \mathrm{J}$

**Q21** Two particles having positive charges $+Q$ and $+2\mathrm{Q}$ are fixed at equal distance $\mathrm{x}$ from centre of a conducting sphere having zero net charge and radius $\mathrm{r}$ as shown. Initially the switch $\mathrm{S}$ is open. After the switch $\mathrm{S}$ is closed, the net charge flowing out of sphere is




(A) $\frac{\mathrm{Qr}}{\mathrm{x}}$
(B) $-\frac{\mathrm{Qr}}{\mathrm{x}}$
(C) $\frac{3Qr}{x}$
(D) $-\frac{3Qr}{x}$

**Q22** Three concentric spherical metallic shells $A, B$ and $C$ of radii $a, b$ and $c(a<b<c)$ have charge densities $\sigma, -\sigma$ and $\sigma$, respectively. If the shells $A$ and $C$ are at the same potential then the correct relation between $a, b$ and $c$ is:

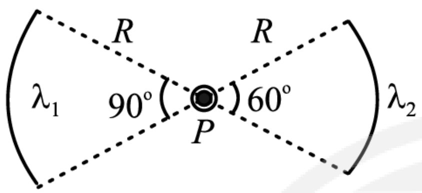
(A) $a+b+c=0$

(B) $a+c=b$

(C) $a+b=c$

(D) $a=b+c$

**Q23** If electric field at point *P* is zero. The find $\frac{\lambda_1}{\lambda_2}$



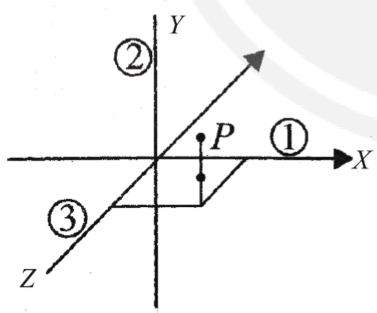

(A) $\frac{\lambda_1}{\lambda_2}=\frac{2}{\sqrt{3}}$

(B) $\frac{\lambda_1}{\lambda_2}=\frac{1}{\sqrt{2}}$

(C) $\frac{\lambda_1}{\lambda_2}=\frac{\sqrt{2}}{3}$

(D) $\frac{\lambda_1}{\lambda_2}=\frac{\sqrt{2}}{1}$

**Q24** Find the electric field vector at *P* (*a*, *a*, *a*) due to three infinitely long lines of charges along the *x*-, *y*- and *z*-axes, respectively. The charge density, i.e., charge per unit length of each wire is $\lambda$



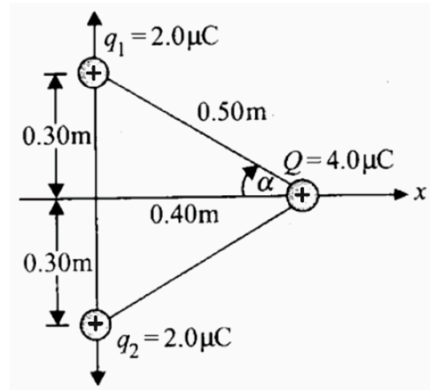

(A) $\frac{\lambda}{3\pi\varepsilon_0 a}\left(\hat{i}+\hat{j}+\hat{k}\right)$

(B) $\frac{\lambda}{2\pi\varepsilon_0 a}\left(\hat{i}+\hat{j}+\hat{k}\right)$

(C) $\frac{\lambda}{2\sqrt{2}\pi\varepsilon_0 a}\left(\hat{i}+\hat{j}+\hat{k}\right)$

(D) $\frac{\sqrt{2}\lambda}{\pi\varepsilon_0 a}\left(\hat{i}+\hat{j}+\hat{k}\right)$

**Q25** In the figure, two equal positive point charges $q_1 = q_2 = 2.0\ \mu C$. The magnitude as well as direction of the net force on *Q* is




(A) 0.23 N in +*x* direction

(B) 0.46 N in +*x* direction

(C) 0.23 N in −*x* direction

(D) 0.46 N in −*x* direction

**Q26** Let $\rho(r)=\rho_0 r^2$ be the charge distribution for a solid sphere of radius *R*, where *r* is the distance from the centre. The magnitude of Electric field at a point which is at the distance of 3*R* from the centre of the sphere is $\frac{\rho_0 R^3}{n\varepsilon_0}$. Find *n*.

**Q27** Two uniformly charged infinitely large plane sheets $S_1$ and $S_2$ are held in air parallel to each other with separation *d* between them. The sheets have charge distributions per unit area $\sigma_1$ and $\sigma_2$($Cm^{-2}$), respectively, with $\sigma_1 > \sigma_2$. The work done by the electric field on a point charge *Q* that moves from $S_1$ towards $S_2$ along a line of length *a* (*a* > *d*) making an angle of π/4 with the normal to the sheets is $\frac{(\sigma_1-\sigma_2)Qa}{y\sqrt{2}\varepsilon_0}$. Find *y*.
(Assume that the charge *Q* does not affect the charge distributions of the sheets.)

**Q28** A stream of a positively charged particles having $\frac{q}{m}=2\times10^{11}\frac{C}{kg}$ and velocity $\vec{v}_0=3\times10^7\hat{i}\,m/s$ is deflected by an electric field $1.8\hat{j}\ \mathrm{kV/m}$. The electric field exists in a region of 10 cm along *x* direction. Due to the electric field, the deflection of the charge particles in the $y$ direction is______ mm.

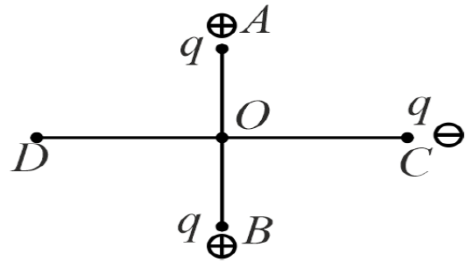
**Q29** Two circular loops of radii $0.05 \mathrm{~m}$ and $0.09 \mathrm{~m}$, respectively are placed such that their axis coincides and their centres are $0.12 \mathrm{~m}$ apart. Charge of $10^{-6}$ coulomb is non-uniformly spread on each of loop such that charge on upper half is twice of that on the lower half. If the potential difference between centres of loop is $N \frac{8}{13} \mathrm{kV}$, then find the value of $N$.

**Q30** Three point charges of magnitude $5 \mu C, 0.16 \mu C$, and $0.3 \mu C$ are located at the vertices $A, B, C$ of a right angled triangle whose sides are $AB=3 \mathit{cm}, BC=3 \sqrt{2} \mathit{cm}$ and $CA=3 \mathit{cm}$ and point $A$ is the right angle corner. Charge at point $A$ experiences__________ $N$ of electrostatic force due to the other two charges.
[JEE Main 2022]

**Q31** Two isolated metallic solid spheres of radii $R$ and $2R$ are charged such that both have same charge density $\sigma$. The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is $\sigma^{\prime}$. The ratio $\frac{\sigma^{\prime}}{\sigma}$ is:[JEE Main 2023, 30 Jan (Shift 1)]
(A) $\frac{9}{4}$
(B) $\frac{4}{3}$
(C) $\frac{5}{3}$
(D) $\frac{5}{6}$

**Q32** A uniformly charged non conducting solid sphere of radius $R$ has potential $V_0$ (measured with respect to $\infty$ ) on its surface. For this sphere the equipotential surfaces with potentials $\frac{3 \mathrm{~V}_0}{2}, \frac{5 \mathrm{~V}_0}{4}, \frac{3 \mathrm{~V}_0}{4}$ and $\frac{\mathrm{V}_0}{4}$ have radius $\mathrm{R}_1, \mathrm{R}_2, \mathrm{R}_3$ and $\mathrm{R}_4$ respectively. Then
(A) $\mathrm{R}_1=0$ and $\mathrm{R}_2<(\mathrm{R}_4-\mathrm{R}_3)$
(B) $2\mathrm{R}>\mathrm{R}_4$
(C) $\mathrm{R}_1=0$ and $\mathrm{R}_2>(\mathrm{R}_4-\mathrm{R}_3)$
(D) $\mathrm{R}_1 \neq 0$ and $(\mathrm{R}_2-\mathrm{R}_1)>(\mathrm{R}_4-\mathrm{R}_3)$

**Q33** Two fixed equal positive charges, each of magnitude $q=5 \times 10^{-5} C$ are located at points *A* and *B*, where *AB* is 6*m*. An equal and opposite charge moves towards *AB* line along the perpendicular bisector of *AB*




The moving charge, when reaches at point *C* at a distance of 4 m from O , has a kinetic energy of 4J. Calculate the distance of farthest point *D* which the negative charge will reach before turning towards *C*.
(A) 0.5 m
(B) 9 m
(C) 9.5 m
(D) 8.5 m

**Q34** A uniformly charged and infinitely long wire having a linear charge density $\lambda$ is placed at a normal distance *y* from a point $\mathrm{O}$. Consider a sphere of radius $\mathrm{R}$ with $\mathrm{O}$ as centre and $\mathrm{R}>\mathrm{y}$. Electric flux through the surface of the sphere is:
(A) Zero
(B) $\frac{2 \lambda R}{\varepsilon_0}$
(C) $\frac{2 \lambda \sqrt{\mathrm{R}^2-\mathrm{y}^2}}{\varepsilon_0}$
(D) $\frac{\lambda \sqrt{\mathrm{R}^2+\mathrm{y}^2}}{\varepsilon_0}$

**Q35** Two conducting spheres of radii $r_1$ and $r_2$ have same electric field near their surfaces. The ratio of their electrical potential is:
(A) $r_1^2 / r_2^2$
(B) $r_2^2 / r_1^2$
(C) $r_1 / r_2$
(D) $r_2 / r_1$

**Q36** A long wire with linear charge density $+\lambda$ is lying along the x-axis. Short electric dipole of dipole moment $\overrightarrow{P}$ is placed at co-ordinate $(x, y, z)$ then

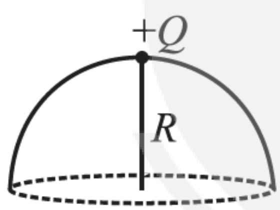
(A) If $\overrightarrow{P} = P_0 \hat{i}$ & dipole is placed at $x = a, y = a, z = a$ then net force will be zero and torque will be non-zero

(B) If $\overrightarrow{P} = P_0 \frac{(\hat{i}+\hat{j})}{\sqrt{2}}$ and dipole is placed at $x = 0$, $y = a$, $z = a$ then potential energy of dipole is zero

(C) If $\overrightarrow{P} = P_0 \frac{(\hat{i}+\hat{j})}{\sqrt{2}}$ and dipole is placed at $x = a$, $y = a$, $z = a$ then potential energy of dipole is zero

(D) If $\overrightarrow{P} = P_0 \hat{j}$ & dipole is placed at $x = a$, $y = a$ & $z = a$ then value of force on dipole is $\frac{1}{4\pi\varepsilon_0} \frac{P_0 \lambda}{\sqrt{2} a^2}$

**Q37** A point charge $+Q$ is placed just outside an imaginary hemispherical surface of radius $R$ as shown in the figure. Which of the following statements is/are correct?



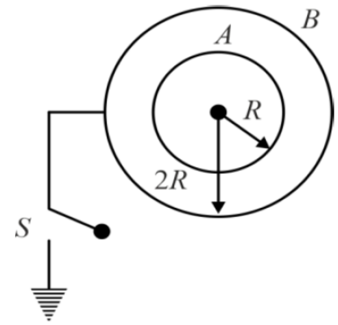

(A) The electric flux passing through the curved surface of the hemisphere is $-\frac{Q}{2\varepsilon_0}\left(1 - \frac{1}{\sqrt{2}}\right)$

(B) Total flux through the curved and the flat surfaces is $\frac{Q}{\varepsilon_0}$

(C) The component of the electric field normal to the flat surface is constant over the surface

(D) The circumference of the flat surface is an equipotential

**Q38** Two concentric conducting spherical shells of radii $R$ and $2R$ having charges $q_A$ and $q_B$ and potential $2V$ and $3V/2$ respectively. Now shell B is earthed by closing switch S . Let charges on spherical shells $A$ and $B$ become $q'_A$ and $q'_B$ respectively, then



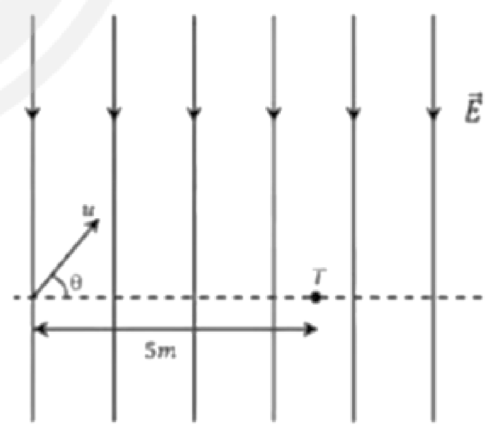

(A) $\frac{q_A}{q_B} = \frac{1}{2}$ (magnitude).

(B) $\frac{q'_A}{q'_B} = 1$ (magnitude).

(C) potential of $A$ after earthing becomes $\frac{3V}{2}$

(D) potential difference between $A$ and $B$ $(V_A - V_B)$ after earthing becomes $\frac{V}{2}$

**Q39** A uniform electric field, $\overrightarrow{\mathrm{E}} = -400\sqrt{3}\hat{y}\mathrm{NC}^{-1}$ is applied in a region. A charged particle of mass $m$ carrying positive charge q is projected in this egion with an initial speed of $2\sqrt{10} \times 10^6 \mathrm{~ms}^{-1}$. This particle is aimed to hit a target $T$, which is $5 \mathrm{~m}$ away from its entry point into the field as shown schematically in the figure. Take $\frac{\mathrm{q}}{\mathrm{m}} = 10^{10} \mathrm{Ckg}^{-1}$. Then




(A) The particle will hit if projected at an angle $45^\circ$ from the horizontal

(B) The particle will hit $\mathrm{T}$ if projected either at an angle $30^\circ$ or $60^\circ$ from the horizontal

(C)

Time taken by the particle to hit T could be $\sqrt{\frac{5}{6}}\,\mu s$ as well as $\sqrt{\frac{5}{2}}\,\mu s$

(D) Time taken by the particle to hit T is $\sqrt{\frac{5}{3}}\,\mu s$

**Q40** Eight point charges each of magnitude *q* are located on the corners of a cube of edge *a*. Then,

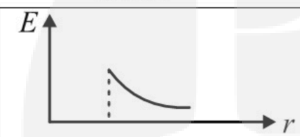
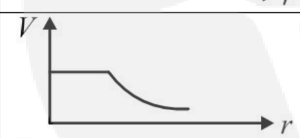
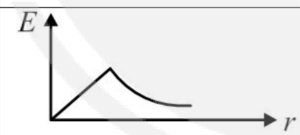
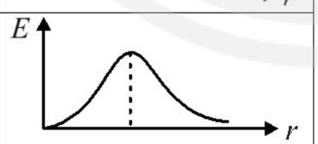
(A) Force on each corner charge due to other charges is 0 N

(B) Force on a charge placed at centre of cube is $\frac{8kq^2}{\left(\frac{\sqrt{3}}{2}a\right)^2}$N

(C) Electric field at centre of top face is $\frac{2^{7/2}}{3^{3/2}}\left(\frac{kq}{a^2}\right)$

(D) Electric field at centre of top face is directed along area vector of the face

**Q41** Match the following two List.

| List-I | | List-II | |
|---|---|---|---|
| I | Electric field due to charged spherical shell | P |  |
| II | Electric potential due to charged spherical shell | Q |  |
| III | Electric field due to charged solid sphere | R |  |
| IV | Electric potential due to charged solid sphere | S |  |
| | | T | None of these |

(A) I→P; II→Q; III→R; IV→T

(B) I→Q; II→P; III→R; IV→T

(C) I→P; II→Q; III→R; IV→S

(D) I→Q; II→P; III→R; IV→S

**Q42** Match List I with List II (Symbols have usual meaning)

| | List-I | | List-II |
|---|---|---|---|
| I | Electric field due to uniformly charged infinite plane insulating thin sheet | P | 0 |
| II | Electric field due to uniformly charged infinite plane conducting sheet of uniform thickness | Q | $\frac{\sigma}{2\varepsilon_0}$ |
| III | Electric field due to nonconducting uniformly charged solid sphere at its surface | R | $\frac{R\rho}{3\varepsilon_0}$ |
| IV | Electric field due to conducting charged solid sphere at its centre | S | $\frac{\sigma}{\varepsilon_0}$ |
| | | T | $\frac{\rho}{3\varepsilon_0}$ |

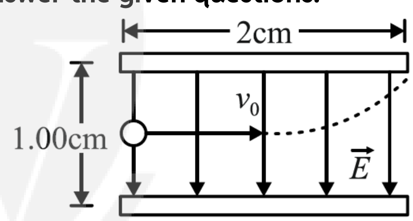
(A) I→Q; II→S; III→R; IV→P

(B) I→R; II→T; III→P; IV→S

(C) I→P; II→T; III→S; IV→R

(D) I→Q; II→P; III→S; IV→R

**Directions (43-44) Read the following passage and answer the given questions.**



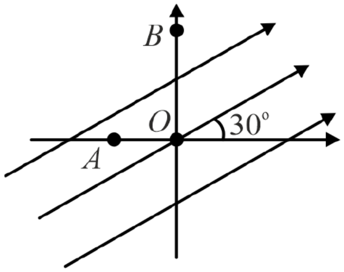

An electron is projected with an initial speed $v_0$ = 1.60 × $10^6$ $ms^{-1}$ into the uniform field between the parallel plates as shown in figure. Assume that the field between the plates is uniform and directed vertically downwards, and that the field outside the plates is zero. The electron enters the field at a point midway between the plates. Mass of electron = 9.1 × $10^{-31}$ kg.

**Q43** If the electron just misses the upper plate, the time of flight of electron up to this instant is $x \times 10^{-8}$ s. Find the value of *x*.

**Q44** For condition of previous situation, the magnitude of electric field (in N/C) is

**Directions (45-46) Read the following passage and answer the given questions.**

A uniform electric field of 100 is directed at 30° with the positive *x*-axis as shown in figure. *OA* = 2 m and *OB* = 4 m $(\sqrt{3}=1.732)$



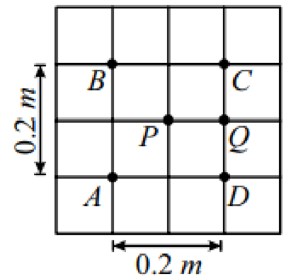

**Q45** The potential difference $V_O - V_A$ (in volts) is
(magnitude only)

**Q46** The potential difference $V_B - V_A$ (in volts) is
(magnitude only)

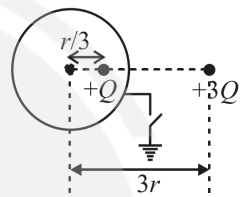
**Q47** Three identical positive charge each *Q* arranged at the vertices of a regular tetrahedron of edge length "*a*". Electric field at the remaining vertex of a regular tetrahedron is $\frac{kxQ}{\sqrt{6}a^2}$; where *x* is

**Q48** Two thin circular rings A and B, both of radius $R$, are fixed coaxially with a distance $\sqrt{3}R$ between their centres and charged uniformly with total charge $+Q$ and $-Q$ respectively. A particle of mass $m$ carrying charge $+q$ is released from the centre of the ring A. If the particle reaches the centre of ring $B$ with velocity $\left(X\left(\frac{Qq}{4\pi\varepsilon_0 Rm}\right)\right)^{1/2}$, then $X$ is ______

**Q49** $A, B, C, D, P$ and $Q$ are points in a uniform electric field. The potentials at these points are $V(A)=2$ volt, $V(P)=V(B)=V(D)=5$ volt and $V(C)=8$ volt. The magnitude of electric field at $P$ is $x\sqrt{2}$ V/m. Find $x$.




**Q50** A conducting shell of radius $r$ has total charge $2Q$ and a point charge $Q$ is at distance $\frac{r}{3}$ from centre as shown. A charge particle having charge $3Q$ is placed outside of shell at distance $3r$ from centre. If the shell is now earthed the amount of charge flown through switch $S$ is $nQ$. Find value of $n$.



# Answer Key

| | | | |
|---|---|---|---|
| Q1 | (D) | Q26 | 45.00 |
| Q2 | (A) | Q27 | 2 |
| Q3 | (D) | Q28 | 2 |
| Q4 | (B) | Q29 | 115 |
| Q5 | (B) | Q30 | 17 |
| Q6 | (B) | Q31 | (D) |
| Q7 | (B) | Q32 | (A) |
| Q8 | (B) | Q33 | (D) |
| Q9 | (C) | Q34 | (C) |
| Q10 | (D) | Q35 | (C) |
| Q11 | (A) | Q36 | (A, D) |
| Q12 | (A) | Q37 | (A, D) |
| Q13 | (C) | Q38 | (A, B, D) |
| Q14 | (A) | Q39 | (B, C) |
| Q15 | (B) | Q40 | (C, D) |
| Q16 | (C) | Q41 | (A) |
| Q17 | (B) | Q42 | (A) |
| Q18 | (B) | Q43 | 1.25 |
| Q19 | (A) | Q44 | 364.00 |
| Q20 | (D) | Q45 | -173.20 |
| Q21 | (C) | Q46 | -373.20 |
| Q22 | (C) | Q47 | 6 |
| Q23 | (B) | Q48 | 2 |
| Q24 | (B) | Q49 | 15 |
| Q25 | (B) | Q50 | 4 |